

FORESIGHT – DEMAND & INVENTORY INTELLIGENCE

PROJECT DOCUMENTATION REPORT

Data Science & Analytics Internship Project

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1. PROJECT OVERVIEW

FORESIGHT is a Demand & Inventory Intelligence project developed as part of a Data Science & Analytics internship.

The main purpose of this project is to use historical sales and inventory data to understand demand patterns, forecast future SKU-level demand, identify inventory risks, and provide useful business recommendations.

The project combines data analysis, machine learning, inventory risk analysis, and an interactive Streamlit dashboard.

The complete workflow includes:

Data Preparation → Exploratory Data Analysis → Demand Forecasting → Forecast Evaluation → Inventory Risk Analysis → Business Dashboard

The project was developed using Python and Jupyter Notebook. Pandas was used for data preparation and analysis, while Scikit-learn was used for the machine learning model. Streamlit was used to create the final interactive dashboard.

2. BUSINESS PROBLEM

Inventory management is an important part of retail business operations.

If demand is higher than the available inventory, products may become unavailable to customers. This creates a possible stockout situation.

On the other hand, if inventory is much higher than expected demand, products may remain unsold for a long period. This creates an overstock situation.

The FORESIGHT project addresses these two problems by using historical sales data and current inventory information.

The project focuses on answering the following questions:

1. How much demand can be expected for each SKU?
 2. Which products may face stockout risk?
 3. Which products may have excess inventory?
 4. What action can be considered for each risk?
 5. What is the estimated inventory value associated with the identified risks?
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3. PROJECT OBJECTIVES

The main objectives of the FORESIGHT project are:

- To prepare and clean the available sales and inventory datasets.
- To understand historical sales patterns.
- To perform exploratory data analysis.
- To create weekly SKU-level demand data.
- To build a seasonal-naive forecasting baseline.
- To build a Random Forest demand forecasting model.
- To evaluate the forecasting model using WAPE.

- To use chronological rolling-origin validation.
 - To identify stockout and overstock risks.
 - To calculate estimated inventory value at risk.
 - To provide recommended actions for different risk conditions.
 - To present the results through an interactive Streamlit dashboard.
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4. TECHNOLOGY STACK

Technology	Purpose
Python	Main programming language
Pandas	Data cleaning and data analysis
NumPy	Numerical operations
Scikit-learn	Machine learning
Random Forest Regressor	Demand forecasting
Matplotlib / Plotting	Data visualization
Jupyter Notebook	Analysis and model development
Streamlit	Interactive dashboard
Git / GitHub	Project version control
Microsoft Word	Project documentation

5. DATASET DESCRIPTION

The project uses four main datasets.

Dataset	Records	Purpose
sales_daily.csv	36,550	Daily sales information
sku_master(1).csv	50	Product and SKU information
calendar.csv	731	Calendar and seasonal information
inventory_snapshots.csv	4,800	Inventory and stock information

5.1 Sales Dataset

The sales dataset contains daily sales information.

Important columns include:

Column	Description
Date	Sales date
SKU	Product identifier
Units_Sold	Number of units sold

Column	Description
Revenue	Sales revenue
Price	Product price
Promotion	Promotion indicator

5.2 SKU Master Dataset

The SKU master contains product-level information.

Important columns include:

Column	Description
SKU	Product identifier
Product_Name	Product name
Category	Product category
Subcategory	Product subcategory
Launch_Date	Product launch date
Cost_Price	Product cost
Selling_Price	Selling price
Gross_Margin_Per_Unit	Gross margin per unit

5.3 Calendar Dataset

The calendar dataset provides time-related information such as:

- Year
- Month
- Quarter
- Week
- Day of week
- Weekend indicator
- Season
- Holiday
- Promotion event

5.4 Inventory Dataset

The inventory dataset contains information required for inventory risk analysis.

Important columns include:

Column	Description
Snapshot_Date	Inventory snapshot date
SKU	Product identifier

Column	Description
Current_Stock	Current available stock
On_Order	Stock already ordered
Lead_Time_Days	Expected replenishment lead time
Safety_Stock	Required safety stock
Reorder_Point	Reorder level
Inventory_Value	Inventory value

6. DATA PREPARATION

The first stage of the project was data preparation.

The datasets were inspected for:

- Data types
- Missing values
- Duplicate records
- Date formats
- Numerical values
- Relationships between datasets

6.1 Data Quality Summary

Dataset	Missing Values	Duplicate Records
Sales	No missing values	0
SKU Master	No missing values	0
Calendar	Missing values in holiday and promotion_event	0
Inventory	No missing values	0

The date columns were converted into proper datetime format before further analysis.

The following date columns were converted:

- Sales Date
- Launch Date
- Calendar date
- Inventory Snapshot Date

6.2 Derived Columns

Additional columns were created for analysis.

These included:

- Revenue Per Unit
- Profit Per Unit
- Total Profit
- Profit Margin Percentage

Revenue per unit was calculated using revenue and units sold.

Where units sold were zero, revenue per unit could not be calculated normally. These resulting missing values were handled before continuing with the analysis.

6.3 Weekly Sales Data

For forecasting, the daily sales data was converted into weekly SKU-level demand.

The weekly data was created by grouping:

Date + SKU

and calculating the total:

Units Sold

This weekly dataset was then used for demand forecasting.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the sales behaviour of the business.

7.1 Overall Sales Information

Metric	Result
Total Sales Records	36,550
Number of Project SKUs	50
Total Project Revenue	₹3,096,056,707

The analysis showed differences in revenue across product categories, years, products, days of the week and promotion periods.

7.2 Revenue by Category

Category	Revenue
Home Decor	₹888.47 million
Furniture	₹603.39 million

Category	Revenue
Storage	₹581.93 million
Kitchen	₹563.53 million
Lighting	₹458.74 million

Home Decor recorded the highest revenue among the categories.

7.3 Revenue by Year

Year	Revenue
2024	₹1.552 billion
2025	₹1.544 billion

The revenue values for the two years were relatively close, with 2024 having the higher total.

7.4 Promotion Analysis

Average units sold during promotion periods were compared with non-promotion periods.

Condition	Average Units Sold
Promotion	18.61
No Promotion	13.48

The average units sold during promotion periods were approximately **38.12% higher** than during non-promotion periods.

This indicates that promotions are associated with higher sales volume in the analysed data.

7.5 Top Products by Revenue

Rank	SKU	Revenue
1	SKU026	₹180.89 million
2	SKU035	₹169.16 million
3	SKU012	₹167.40 million
4	SKU042	₹152.85 million
5	SKU027	₹150.89 million

7.6 Monthly Sales Observation

March recorded the highest monthly revenue in both 2024 and 2025.

This indicates that certain months may have stronger demand and should be considered when planning future forecasting improvements.

8. DEMAND FORECASTING

Demand forecasting is one of the main components of the FORESIGHT project.

The objective was to estimate future weekly demand for each SKU.

8.1 Weekly Demand

The sales data was grouped by week and SKU.

For every SKU, weekly Units_Sold was calculated.

The resulting weekly data covered approximately 105 weeks.

8.2 Seasonal-Naive Baseline

A seasonal-naive forecasting method was used as the baseline.

The forecast for a week was based on the demand from four weeks earlier.

This provided a simple seasonal reference against which the machine learning model could be evaluated.

The seasonal-naive baseline achieved:

WAPE = 13.36%

8.3 Machine Learning Model

A Random Forest Regressor was selected for demand forecasting.

The final model used the following features:

Feature	Description
Baseline_Forecast	Previous-week demand baseline
Lag_2	Demand from two weeks earlier
Lag_3	Demand from three weeks earlier
Rolling_4_Weeks	Average demand from the previous four weeks

The rolling feature was created using previous observations only.

This was important because using the current week's demand while predicting that same week's demand would introduce target leakage.

8.4 Final Model

The final Random Forest model used:

Parameter	Value
Model	Random Forest Regressor
Number of Trees	100
Random State	42
Training Observations	5,050

The model was trained using the complete leakage-free training dataset.

9. FORECAST EVALUATION

Forecast performance was evaluated using **WAPE – Weighted Absolute Percentage Error**.

WAPE was selected because it measures the total forecasting error relative to the total actual demand.

A chronological rolling-origin validation approach was used.

Random train-test splitting was avoided because future observations should not be used to predict the past.

9.1 Rolling-Origin Validation Results

Validation Fold	WAPE
Fold 1	15.39%
Fold 2	11.27%
Fold 3	9.78%
Average	12.15%

9.2 Baseline Comparison

Method	WAPE
Seasonal-Naive Baseline	13.36%
Random Forest Model	12.15%

The Random Forest model achieved an average WAPE of **12.15%**, compared with **13.36%** for the seasonal-naive baseline.

The relative improvement was approximately:

9.06%

This result is based on the leakage-free rolling-origin evaluation.

10. INVENTORY RISK ANALYSIS

The next stage was to combine demand forecasting with inventory information.

The final analysis was performed for the 50 project SKUs.

The latest inventory snapshot used in the analysis was:

1 December 2025

10.1 Next-Week Forecast

A next-week demand forecast was generated for each project SKU.

The forecast was then used as an input for inventory risk calculations.

10.2 Stockout Risk

Stockout risk was calculated using:

Daily Forecast Demand = Next-Week Forecast / 7

Then:

Lead-Time Demand = Daily Forecast Demand × Lead Time Days

Available stock was calculated as:

Available Stock = Current Stock + On Order

Projected stock after lead time was calculated as:

Projected Stock = Available Stock – Lead-Time Demand

A SKU was classified as having stockout risk when:

Projected Stock < Safety Stock

This identifies products where expected demand during replenishment lead time may reduce inventory below the required safety level.

10.3 Overstock Risk

Four-week demand was estimated using the next-week forecast.

The calculation used:

$$\text{Four-Week Demand} = \text{Next-Week Forecast} \times 4$$

A SKU was classified as overstock risk when:

$$\text{Current Stock} > \text{Four-Week Demand} \times 2$$

This identifies products where current inventory is significantly higher than the estimated near-term demand.

10.4 Risk Levels

Four risk conditions were used.

Risk Level	Meaning	Recommended Action
Healthy	No identified stockout or overstock risk	Healthy
High Stockout	Possible stock shortage	Reorder Now
High Overstock	Inventory significantly above expected demand	Markdown / Clear
High Both	Both stockout and overstock conditions	Watch / Volatile

10.5 Final Risk Results

Risk Level	Number of SKUs
Healthy	32
High Overstock	13
High Stockout	5
High Both	0
Total	50

Therefore, 18 of the 50 project SKUs were identified with either high stockout or high overstock risk.

10.6 Estimated Inventory Value at Risk

The project calculated an estimated inventory value associated with the identified risks.

Total Estimated Value at Risk: ₹16,656,255.78

Approximately:

₹1.67 crore

This value is a model-based estimate generated using the project's risk rules. It should not be interpreted as an actual realized financial loss.

11. BUSINESS RECOMMENDATIONS

Based on the forecasting and inventory risk analysis, the following actions can be considered.

11.1 Stockout Products

The five high-stockout SKUs should be reviewed for replenishment.

The business can consider:

- Reordering inventory.
- Checking supplier lead times.
- Reviewing safety-stock levels.
- Monitoring upcoming promotions that may increase demand.

11.2 Overstock Products

The thirteen high-overstock SKUs should be reviewed for inventory reduction.

Possible actions include:

- Markdown.
- Clearance sales.
- Promotion planning.
- Reducing future replenishment.
- Monitoring slow-moving products.

11.3 Healthy Products

The 32 healthy SKUs can continue to be monitored using the same forecasting and risk process.

11.4 Continuous Monitoring

Forecast accuracy should be monitored as new sales data becomes available.

The forecasting model can also be improved later by including additional features such as:

- Promotion information
- Price changes

- Seasonal patterns
 - Calendar effects
 - Additional lag features
 - Additional forecasting models
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12. STREAMLIT DASHBOARD

The project results were integrated into an interactive Streamlit dashboard.

The dashboard provides a simple interface for viewing the main analytical results.

12.1 Dashboard Pages

Page	Purpose
Home	Overall project KPIs and summary
Sales Analytics	Sales trends and business insights
Demand Forecast	Actual demand and forecast
Inventory Risk	Risk levels and recommended actions
Executive Summary	Main business findings

12.2 Home Page

The Home page displays important project-level KPIs such as:

- Total Revenue
- Number of High-Risk SKUs
- Estimated Value at Risk
- Risk Summary
- Recommended Actions
- Top Products

12.3 Sales Analytics Page

This page provides sales-related analysis such as:

- Revenue trends
- Category performance
- Product performance
- Sales patterns

12.4 Demand Forecast Page

The Demand Forecast page allows the user to select an SKU and view its historical demand and forecast.

For example, the generated next-week forecast for **SKU001** was approximately:

85.68 units

for the forecast week beginning **11 January 2026**.

12.5 Inventory Risk Page

The Inventory Risk page displays:

- Risk level
- Current stock
- Forecast demand
- Stockout risk
- Overstock risk
- Recommended action
- Estimated value at risk

12.6 Executive Summary Page

The Executive Summary presents the major findings in simple business language.

The main actions are:

- Reorder Now
- Markdown / Clear
- Watch / Volatile
- Healthy

13. PROJECT RESULTS

The main final results of the FORESIGHT project are shown below.

Metric	Final Result
Sales records analysed	36,550
Project SKUs	50
Weekly model observations	5,050
Seasonal-naive WAPE	13.36%
Random Forest WAPE	12.15%
Relative improvement	9.06%
Healthy SKUs	32

Metric	Final Result
High Overstock SKUs	13
High Stockout SKUs	5
High Both SKUs	0
Estimated Value at Risk	₹1.67 crore

Key Findings

Finding 1 – Demand Forecasting

The Random Forest model achieved an average WAPE of 12.15% during rolling-origin validation, compared with 13.36% for the seasonal-naive baseline.

Finding 2 – Inventory Risk

Out of 50 project SKUs, 5 were classified as high stockout risk and 13 as high overstock risk.

Finding 3 – Business Value

The estimated inventory value associated with the identified risks was approximately ₹1.67 crore.

Finding 4 – Promotion Impact

Average units sold during promotion periods were approximately 38.12% higher than during non-promotion periods in the analysed data.

Finding 5 – Dashboard

The Streamlit dashboard converts the analytical outputs into business-facing information that can support inventory monitoring and planning.

14. LIMITATIONS

The project has the following limitations:

1. The forecasting model uses a relatively small set of historical demand features.
2. The current forecast focuses on one-week-ahead SKU-level demand.
3. The inventory risk thresholds are rule-based and may need adjustment based on actual business policies.
4. The estimated value at risk is a model-based estimate and should not be considered a realized financial loss.
5. Additional forecasting techniques could be tested in future versions.
6. Model performance may change when new sales and inventory data become available.

15. CONCLUSION

FORESIGHT provides an end-to-end Demand & Inventory Intelligence workflow.

The project begins with data preparation and exploratory data analysis. Weekly SKU-level demand is then created and used for forecasting.

A seasonal-naive method was used as a baseline, followed by a Random Forest machine learning model.

The Random Forest model achieved an average WAPE of **12.15%**, compared with **13.36%** for the seasonal-naive baseline, resulting in an improvement of approximately **9.06%**.

The inventory risk analysis identified:

- **32 Healthy SKUs**
- **13 High Overstock SKUs**
- **5 High Stockout SKUs**
- **0 High Both SKUs**

The estimated inventory value at risk was approximately **₹1.67 crore**.

Finally, the Streamlit dashboard presents the forecasting and inventory results in a simple and interactive format.

Overall, the FORESIGHT project demonstrates how data analysis, machine learning and business rules can be combined to support demand forecasting and inventory planning.

16. APPENDIX – PROJECT FILES

File / Folder	Purpose
notebooks/FORESIGHT_Project.ipynb	Main analysis and modelling notebook
data/	Project datasets
outputs/clean_sales.csv	Cleaned sales data
outputs/sku_master.csv	SKU master output
outputs/weekly_forecast.csv	Weekly forecasting results
outputs/next_week_forecast.csv	Next-week SKU forecasts
outputs/risk_results.csv	Final inventory risk results
app/app.py	Streamlit dashboard application
app/style.css	Dashboard styling
README.md	Project overview and reproducibility information

File / Folder	Purpose
requirements.txt	Python package requirements
models/	Model-related project files
documentation/	Project documentation

END OF REPORT

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